

Achievement 2.5: Visual Applications of Machine Learning

This report explores how visual machine learning techniques can be applied to digit recognition and weather image classification as a proof of concept for ClimateWins.

Part 1: Handwriting Recognition

In this exercise, a convolutional neural network trained on the MNIST dataset was used to recognize handwritten digits. Ten handwritten numbers (0–9) were created using natural handwriting, digitized, resized to 28×28 pixels, normalized, and tested against the trained model.

The model correctly identified **7 out of 10 digits**, resulting in an accuracy of **70%**. The misclassified digits (0, 3, and 8) were clearly written and visually unambiguous to a human observer, indicating that the errors were not due to handwriting quality. Instead, these misclassifications likely resulted from information loss during image resizing and differences between real-world handwritten inputs and the MNIST training data. This highlights a common limitation of pretrained image classifiers when applied outside their original training domain.

File	True	Predicted	Correct
0.jpg	0	9	No
1.jpg	1	1	Yes
2.jpg	2	2	Yes
3.jpg	3	9	No
4.jpg	4	4	Yes
5.jpg	5	5	Yes
6.jpg	6	6	Yes
7.jpg	7	7	Yes
8.jpg	8	3	No
9.jpg	9	9	Yes

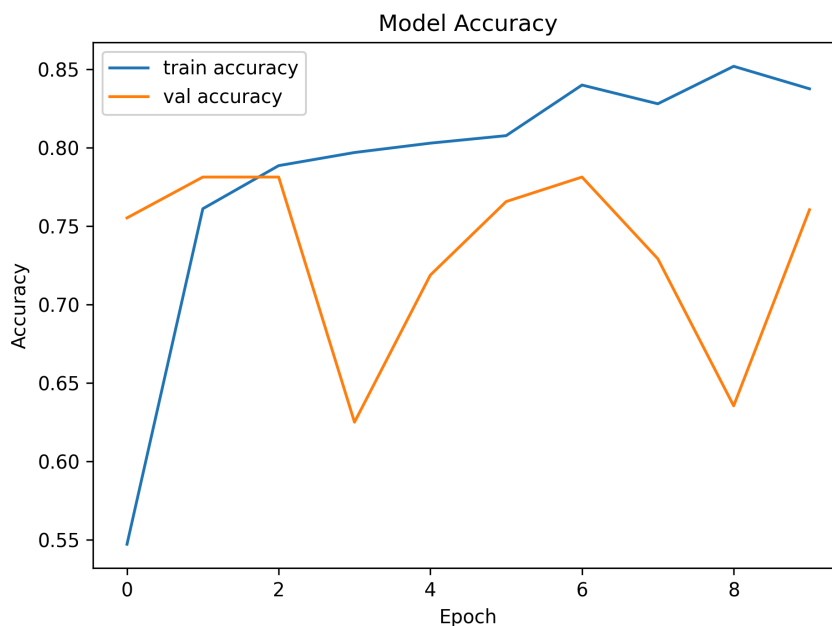
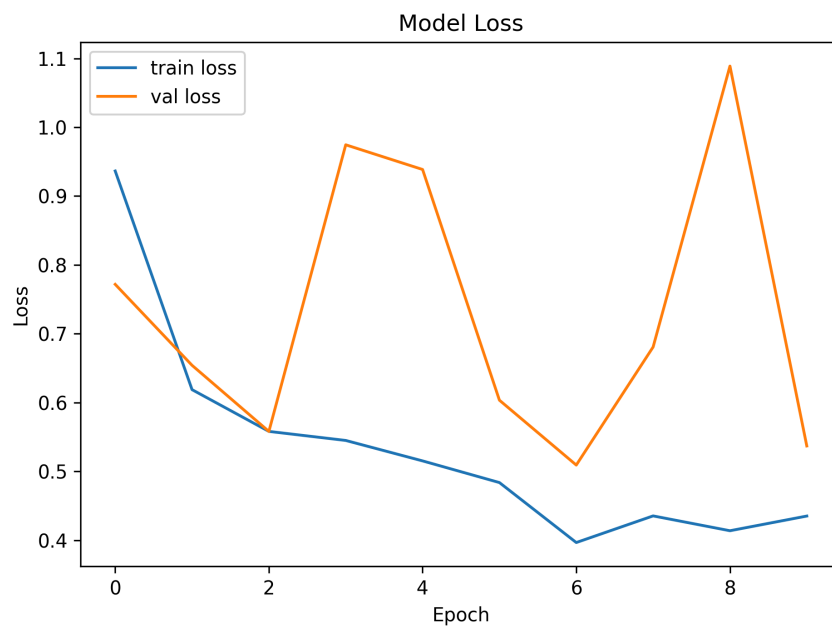
Accuracy: 70.00% (7/10)

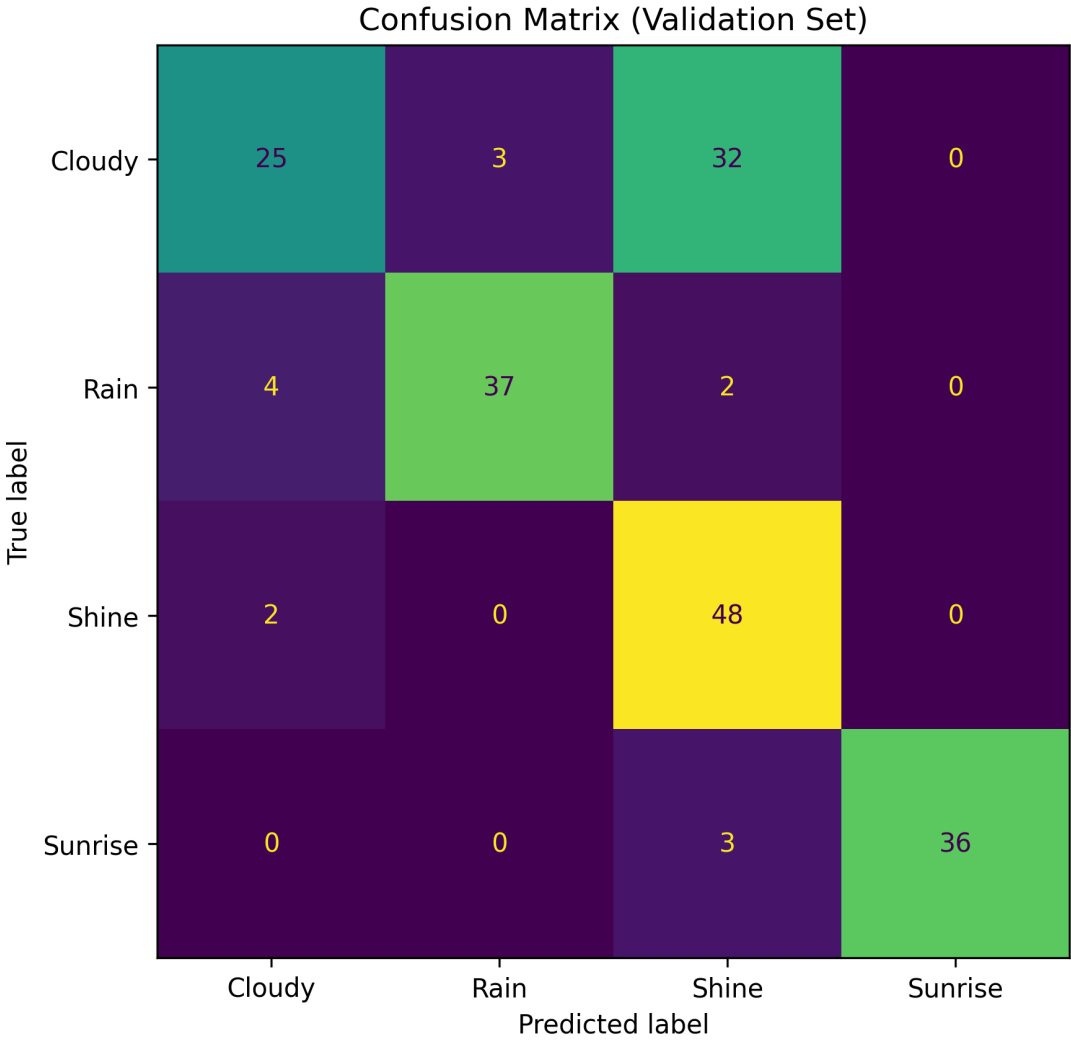
Part 2: Weather Image Recognition

A multi-class convolutional neural network was trained to classify weather conditions (cloudy, rainy, sunny, and sunrise) using a labeled image dataset from Kaggle. Images were rescaled and split into training and validation sets using an 80/20 split. The model architecture consisted of multiple convolutional and max-pooling layers followed by fully connected layers with dropout to reduce overfitting.

The model was trained for **10 epochs**, with convergence observed between **epochs 5 and 7** based on validation loss and accuracy trends. Training beyond this range did not consistently improve validation performance and introduced signs of overfitting. Final validation accuracy stabilized around **75–78%**, with corresponding loss values indicating reasonable generalization to unseen images.

Loss and accuracy curves, along with a confusion matrix, were used to evaluate model performance. The confusion matrix showed that visually similar weather conditions (e.g., cloudy vs. rainy) were more frequently misclassified, reflecting inherent ambiguity in visual-only weather classification. Additionally, examples of how accurate the models performed on selected images were included in the notebook as well.





Model Loss Plot:

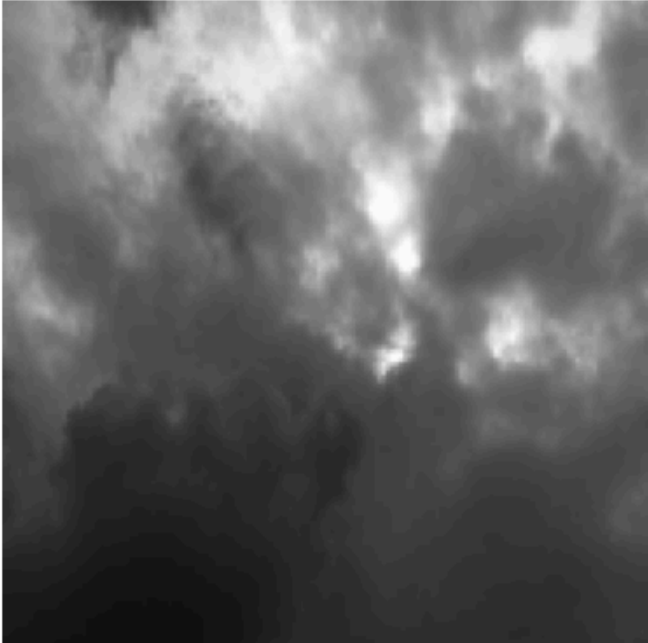
Training loss decreases consistently, whereas validation loss varies substantially across epochs. This suggests sensitivity to unseen data and reinforcing the presence of overfitting beyond the optimal training range.

Model Accuracy Plot:

Training accuracy increases steadily across epochs while validation accuracy fluctuates. This indicates that the model learns meaningful patterns but begins to overfit after early convergence.

Confusion Matrix:

The confusion matrix shows that the model most accurately classifies *Shine* and *Sunrise* conditions, while confusion occurs primarily between *Cloudy* and *Shine*, which shows visual similarity between these categories.

Train Generator Examples	Validation Generator Examples
Correct true: Shine pred: Shine (p=0.97) 	Correct true: Cloudy pred: Cloudy (p=0.80) 
Correct true: Cloudy pred: Cloudy (p=0.76) 	Correct true: Cloudy pred: Cloudy (p=0.65) 

Sample predictions from the training and validation sets illustrate correct classification of distinct weather conditions

Part 3: Generative Adversarial Networks (GANs) for Weather Prediction

GANs offer promising applications in weather prediction with their ability to generate realistic visual data:

1. **Synthetic weather scenario generation**

GANs can be used to generate synthetic radar or satellite imagery for rare or extreme weather events, bridging gaps in limited historical datasets and improving model robustness. Similar approaches have been explored in precipitation nowcasting research by DeepMind.

2. **Super-resolution of weather imagery**

GANs can enhance the spatial resolution of low-resolution weather maps, which enables more detailed forecasts without requiring higher-cost sensing infrastructure. This approach has been investigated in climate and remote sensing research.

3. **Probabilistic forecast visualization**

GANs can generate multiple plausible future weather scenarios from a single input image, which supports uncertainty-aware forecasting and improved decision-making in climate risk management.

<https://arxiv.org/html/2409.11502v1>

<https://www.deeplearning.ai/the-batch/weather-forecast-by-gan/>

Conclusion

This proof-of-concept demonstrates that machine learning models can extract meaningful information from visual data for both digit recognition and weather classification. While visual models perform well under controlled conditions, their limitations highlight the importance of careful preprocessing, domain-specific training data, and uncertainty-aware approaches. GAN-based methods represent a promising extension for generating, enhancing, and interpreting weather-related visual data in future ClimateWins applications.